1. Machine Learning Implementation

Use your knowledge of machine learning to answer the following questions. You could either set up some math (no need to solve) or write pseudocode, your choice (unless otherwise noted)!

A) Use Bayesian inference to calculate the relative likelihood of $p(\theta | \text{data})$ given the following prior probabilities: $p(\theta=0.5) = 0.1$ and $p(\theta=0.25) = 0.2$ and data = Heads, heads, tails, tails, heads and where θ represents likelihood of heads. If you go the pseudocode route, you don't need to calculate each scenario.

B) You have a piece of code (say a function/method) called `SVM` that takes `data` and calculates a model (say, called with `x = SVM(data)`). You also have a function/method called `evaluate` that takes the output of `SVM` and returns an integer with how good the model is. Write a brief bit of pseudocode that performs leave-one-out cross validation.

C) You have the following data, and you'd like to learn "happy" vs "sad" based on the associated variables:
Point A: happy, +1, "cherries"
Point B: happy, -1, "cherries"
Point C: sad, -1, "blueberries"
Develop a tree (hand-drawn) or write pseudocode to build a decision tree as discussed in class (i.e. greedily) with this data:

2. Designing an Algorithm

Design a machine learning algorithm to determine if the weather is going to be "sunny," "partly sunny," "cloudy," "rainy," or "snowy" based on several variables: the predicted high temperature, the predicted low temperature, the historical state (on this day in the past, what was the state?), and the state the previous day. You won't need to implement anything, just make a few choices, as follows:

A) What algorithm would you select to address this problem? Why? What are some of the advantages or disadvantages of your selected method?

B) What parameter(s) would you need to select and either what values would you pick or what method would you use to pick them?

C) How would you check your method for correctness of fit?

3. Selection

Select an algorithm for each of the following scenarios. Please explain WHY you are selecting your algorithm of choice by listing any advantages/disadvantages of your method (i.e. why the advantages are particularly advantageous or why the disadvantages are not too big of a deal).

A) You want to determine if an email is spam or not based on if there has been previous contact, what domain it is from, and the time of day that it was sent. You have a very large amount of training data.

B) You want to predict how acidic soil will be based on global coordinates (longitude and latitude), how "developed" the area is (urban, suburban, rural, or wilderness), and the amount of annual rainfall.

C) You want to predict if a consumer will want to "buy" or "not buy" a video game based on other video games that they've purchased. You have a variable amount training data for each consumer (some will have a lot of training data [i.e., they've purchased a relatively large number of games], while others will have a little). This will be based on the genre, the developer, and the operating system requirements of previous games purchased.

4. Fix it, Felix!

There's something wrong with the following scenarios. Explain what's wrong and fix it. No math or coding is required.

A) You are trying to classify jokes as funny or not funny based on two parameters and you use perceptron to do this because it is very quick.

B) You are using Bayesian inference to determine if a coin is magic or not. You assign prior probability that the coin lands on heads half the time, $p(\theta = 0.5)$, as 0. I mean, the coin's magic, so this makes sense, right?

C) You want to model a home's sale price based on square footage, zip code, year built, and year of most recent remodel. You get this information (as well as the final sale price) in your training data and use linear regression to fit a line. The fit is poor, but you know you have strong training data and never specified any parameters.